

ADITYA SATULURI

Systems Engineer – Grade C1

[LinkedIn](#) — [GitHub](#)

Phone: +91-9208281688

Email: aditya.satuluri@tcs.com

Hyderabad, Telangana, India

TECHNICAL PROJECTS

CodeSight AI – AI-Powered Code Review Summarizer

March 2026

TCS AI Fridays (Runner-Up)

- Rapidly prototyped an AI-driven code review assistant in an **8-hour sprint**, delivering a functional MVP.
- Built a high-velocity **RAG pipeline** using **LangChain**, **ChromaDB**, for **real-time repository analysis**.
- Developed data ingestion and chunking pipelines preserving code structure and metadata.
- Led backend orchestration and vector database integration under strict deadlines.

Wander AI – Agentic Travel Planning Application

May 2026

github.com/adityasatuluri/Travel-Partner

- Engineered a multi-agent travel planner using **LangGraph** and **Flask**, implementing specialized AI nodes (Extractor, Researcher, Planner, Critic) for complex reasoning and **Pydantic** for strict JSON schema validation.
- Architected a dynamic **real-time weather simulator** allowing in-place itinerary replanning with zero-latency DOM updates and interactive visual mismatch highlights.
- Optimized data ingestion by executing 5 external geographic services (**OpenWeatherMap**, **OpenRouteService**, **Geoapify**, **Unsplash**) concurrently via **ThreadPoolExecutor**, and integrated **SQLiteCache** for **semantic LLM caching**, drastically reducing latency.

Human Activity Recognition – Deep Learning Model Performance Analysis

May 2026

[GitHub Repository](#)

- Built a **Human Activity Recognition (HAR)** pipeline using transfer learning with **EfficientNetB3**, **ResNet50**, and **MobileNetV2** to classify **15 human activities**.
 - Implemented staged **fine-tuning** and analyzed convergence behavior using training and validation curves.
 - Achieved highest performance using **EfficientNetB3** with **81% accuracy**, outperforming **ResNet50 (77%)** and **MobileNetV2 (74%)**, showing stronger class-wise prediction quality.
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TECHNICAL SKILLS

GenAI & ML: Agentic AI, RAG, Fine-Tuning, Prompt Engineering, LangChain, Vector Search, NER

Languages: Python, JavaScript, TypeScript

Frontend & Backend: React, Express.js, Node.js, Spring Boot, Angular

Cloud & DevOps: AWS (S3, EC2, Lambda, Glue), Docker, Git

Databases and APIs: ChromaDB, Hibernate, REST APIs, FastAPI

ACHIEVEMENTS & AWARDS

- Runner-Up, **TCS AI Fridays (2026)** — Awarded for CodeSight AI.
 - Runner-Up, **Smart India Hackathon (2023)** — Developed PsyCheck, an AI-based mental health diagnostic application.
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CERTIFICATIONS & COMPETENCIES

- Complete Agentic AI Bootcamp With LangGraph And LangChain — Udemy
- **Generative AI** — **E1** (Training to be E2)
- **Python** — **E1**
- PCAP: Python Essentials — Python Institute
- React Basics — Meta